

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (Bsc.IT Semester II)**  
**Scala for Data Science Practical**

## Practical - 10

|                                |                                     |
|--------------------------------|-------------------------------------|
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| Class :SYIT                    | Batch:01                            |
| Date of Assignment :04/08/2026 | Date/Time of Submission :04/08/2026 |

### Practical 10

Implement linear regression using Breeze. Fit a model to a small dataset and predict a value.

INPUT:

```
Demo.scala x
1  import breeze.linalg._
2  import breeze.stats._
3  import breeze.numerics._
4
5  object Demo extends App {
6    val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0)
7    val y = DenseVector(2.0, 4.1, 6.0, 8.1, 9.9)
8
9    val mX = mean(x)
10   val mY = mean(y)
11
12   val n = sum((x - mX) ** (y - mY))
13   val d = sum((x - mX) ** (x - mX))
14
15   val s = n / d
16   val i = mY - s * mX
17
18   println(f"Slope: $s%.2f")
19   println(f"Intercept: $i%.2f")
20
21   val nX = 6.0
22   val p = s * nX + i
23
24   println(f"Predicted value = $p%.2f")
25
26   println(f"S006 Sejal Deshmukh")
27 }
```

OUTPUT:

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
Slope: 1.98
Intercept: 0.08
Predicted value = 11.96
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Process finished with exit code 0
```

